PERTURBATION THEORY WITH A TRIDIAGONAL ZEROTH-ORDER HAMILTONIAN

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We generalize the Rayleigh–Schrödinger perturbation formalism to the hamiltonians $H=H_0+\lambda H_1$ where the correction λH_1 is small and the unperturbed operator H_0 is represented by an infinite tridiagonal matrix. This enables us to construct the solutions $E=E_0+\lambda E_1+\lambda^2 E_2+...$ and $|\psi\rangle=|\psi_0\rangle+\lambda|\psi_1\rangle+\lambda^2|\psi_2\rangle+...$ in terms of the analytic continued fractions.

A knowledge of the unperturbed states $|n\rangle$ and of the unperturbed hamiltonians

$$H_0=\sum_{n=0}^{\infty}|n\rangle a_n\langle n| \tag{1}$$

is needed in most perturbation theories [1]. Usually, we assume that the operator H_0 is a good approximation to the exact hamiltonian H and solve the full Schrödinger equation

$$H|\psi\rangle=E|\psi\rangle \tag{2}$$

by the power series (Rayleigh–Schrödinger) ansatz

$$E=E_0+\lambda E_1+\lambda^2 E_2+..., \quad |\psi\rangle=|\psi_0\rangle+\lambda|\psi_1\rangle+\lambda^2|\psi_2\rangle+... . \tag{3}$$

Here, the parameter λ characterizes formally the smallness of the perturbation $H-H_0=\lambda H_1$.

The Rayleigh–Schrödinger (RS) approach to (2) may fail to work even for the simplest anharmonic oscillator

$$H=-\Delta+\mu r^2+\nu r^4 , \quad \nu>0 . \tag{4}$$

Indeed, when we interpret νr^4 as a perturbation with $\lambda=\nu$, we find that the RS series (3) diverges [2]. A number of papers (cf., e.g., their list in ref. [3]) has been devoted to a resummation and meaningful re-interpretation of this result. Also, a natural source of convergence has been sought in a repartitioning of H and found in an improved choice of H_0 (1) with $a_n=\langle n|(-\Delta+r^2)^2|n\rangle$ [4], $a_n=\langle n|H|n\rangle$ [5], etc. In all these cases, a guarantee of further acceleration of the slow rate of convergence would still be needed.

In practice, a change of the basis $|n\rangle$ is rarely desirable. For example, the harmonic-oscillator states are easily generalized to the many-body case, their polynomial structure simplifies an evaluation of the matrix elements of H_1, etc. Hence, we encounter a challenge to generalize H_0 (1) to a more general class of matrices which are not entirely diagonal.

In the present letter, we shall consider the tridiagonal unperturbed hamiltonians

$$H_0=\sum_{n=0}^{\infty}|n\rangle a_n\langle n|+\sum_{n=0}^{\infty}|n\rangle b_n\langle n+1|+\sum_{n=0}^{\infty}|n+1\rangle b_{n-1}\langle n| . \tag{5}$$

We shall describe the corresponding extension of the RS formalism and show that a lot of its merits may still be preserved. Mathematically our use of the analytic continued fractions [6] will in effect enable us to compress the redundant set of parameters a_n and b_n into a single auxiliary sequence again.

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The easiest way to proceed is to parallel simply the textbook presentation of the RS theory. Thus, inserting (3) into (2),

$$[H_0-E_0+\lambda(H_1-E_1)-\lambda^2E_2-...](|\psi_0\rangle+\lambda|\psi_1\rangle+...)=0$$

(to be satisfied for each λ) we arrive at a sequence of the independent requirements

$$H_0|\psi_0\rangle=E_0|\psi_0\rangle \tag{6}$$

and

$$(H_0-E_0)|\psi_k\rangle=-H_1|\psi_{k-1}\rangle+\sum_{i=1}^{k}E_i|\psi_{k-i}\rangle\,,\quad k=1,2,... . \tag{7}$$

When we multiply (7) by $\langle\psi_0|$ from the left, we obtain the well-known RS recurrent definition of energies,

$$E_k=\frac{1}{\langle\psi_0|\psi_0\rangle}\left(\langle\psi_0|H_1|\psi_{k-1}\rangle-\sum_{j=1}^{k-1}E_{k-j}\langle\psi_0|\psi_j\rangle\right),\quad k=1,2,... . \tag{8}$$

Thus, we may treat E_k as mere abbreviations and re-interpret (6) and (7) as an implicit recurrent definition of the wavefunction components.

Immediately, we may notice that the overlaps

$$\alpha_k=\langle\psi_0|\psi_k\rangle=\langle\psi_0|0\rangle\langle 0|\psi_k\rangle+\sum_{m=1}^{\infty}\langle\psi_0|m\rangle\langle m|\psi_k\rangle\,,\quad k=1,2,... \tag{9}$$

remain unspecified by (6) and (7). We shall choose here the free parameters α_k in such a way that

$$\langle 0|\psi_k\rangle=\frac{1}{\langle\psi_0|0\rangle}\left(\alpha_k-\sum_{m=1}^{\infty}\langle\psi_0|m\rangle\langle m|\psi_k\rangle\right)=0\,,\quad k=1,2,... . \tag{10}$$

This differs from the RS theory where $\alpha_k^{(RS)}=\langle\psi_0|\psi_k^{(RS)}\rangle=0$, so that we have now $|\psi_k\rangle=|\psi_k^{(RS)}\rangle+\alpha_k|\psi_0\rangle\langle\psi_0|\psi_0\rangle^{-1}$.

In the next step, multiplication of (7) by $\langle n|$,

$$\sum_{m=1}^{\infty}\langle n|(H_0-E_0)|m\rangle\langle m|\psi_k\rangle=...\,,$$

gives

$$|\psi_k\rangle=R_0(E_0)\left(\sum_{i=1}^{k}E_i|\psi_{k-i}\rangle-H_1|\psi_{k-1}\rangle\right),$$

$$R_0(E_0)=Q[Q(H_0-E_0)Q]^{-1}Q\,,\quad k=1,2,...\,, \tag{11}$$

where $Q=1-|0\rangle\langle 0|$. This is the final explicit formula for the wavefunctions.

In (6)–(11), an arbitrary matrix H_0 may be used in principle. Although the RS theory seems to be the only formalism inverting $Q(H_0-E_0)Q$ in a non-numerical manner, an alternative construction of the unperturbed propagator $R_0(E_0)=[Q(H_0-E_0)Q]^{-1}$ is possible also by means of analytic continued fractions [7]. This is our central idea. Its four essential non-RS technical features may be characterized as follows.

(1) *The continued fractional formula for* $R_0(E_0)$. We shall transform the parameters E_0, a_n and b_n into a new auxiliary sequence of quantities $f_n(E_0)$. They bare defined by the recurrences

$$f_k=(a_k-E_0-b_k^2f_{k+1})^{-1}\,,\quad k=1,2,...N\,,\quad f_{N+1}=0\,, \tag{12}$$

in the limit $N\to\infty$, i.e., as the analytic continued fractions [6]. Then, with the abbreviation

$$d_k = \prod_{m=1}^{k} (-b_m f_{m+1}) , \quad d_0 = 1 ,$$

it is easy to show that $R_0(E_0)$ is a simple matrix,

$$Q \frac{1}{QH_0Q - E_0Q} Q = \begin{pmatrix} 1 & & \\ d_1 & 1 & \\ d_2 & d_1 & 1 \\ \ldots & & \end{pmatrix} \begin{pmatrix} f_1 & & & \\ & f_2 & & \\ & & f_3 & \\ & & & \ddots \end{pmatrix} \begin{pmatrix} 1 & d_1 & d_2 & d_3 & \ldots \\ & 1 & d_1 & d_2 & \\ & & 1 & d_1 & \\ & & & \ldots & \end{pmatrix} . \tag{13}$$

Its insertion into (11) leads to the generalized RS formula

$$\langle m|\psi_k\rangle = \sum_{i=1}^{m-1} d_{m-i-1} f_{i+1} \sum_{n=i+1}^{\infty} d_{n-i-1}$$

$$\times \left(-b_{n-1}\langle n-1|\psi_{k-1}\rangle - a_n\langle n|\psi_{k-1}\rangle - b_n\langle n+1|\psi_{k-1}\rangle + \sum_{j=1}^{k} \langle n|\psi_{k-j}\rangle E_j \right) ,$$

$$m,\ k = 1,2,\ldots , \tag{14}$$

containing the continued fractions. It is reducible precisely to the standard RS prescription when we put $b_{m-1}=0$, i.e., $f_m = (a_m - E_0)^{-1}$ and $d_m = 0$ for all $m \geqslant 1$.

(2) *The continued fractional formula for* $|\psi_0\rangle$. The unperturbed equation (6) is closely related to the preceding construction. In accord with ref. [8], we may write

$$\langle m|\psi_0\rangle = g_m = -g_0 b_0 f_1 d_{m-1} , \quad m = 1,2,\ldots , \tag{15}$$

and convert also the RS energy (8) into the continued fractional formula

$$E_1 = \mathcal{N} \sum_{m,n=0}^{\infty} g_m g_n \langle m|H_1|n\rangle , \quad \mathcal{N} = \left(\sum_{i=0}^{\infty} g_i^2 \right)^{-1} ,$$

$$E_k = \mathcal{N} \sum_{n=1}^{\infty} \sum_{m=0}^{\infty} g_m \langle m|H_1|n\rangle \langle n|\psi_{k-1}\rangle - \mathcal{N} \sum_{n=1}^{\infty} \sum_{j=1}^{k-1} E_{k-j} g_n \langle n|\psi_j\rangle , \quad k = 2,3,\ldots . \tag{16}$$

Again, it degenerates to the ordinary RS prescription for $b_{m-1}=0$ since $g_m = 0$, $m \geqslant 1$.

(3) *The continued-fractional determination of* E_0 *or* a_0. The unperturbed energy E_0 should be an eigenvalue of the unperturbed matrix H_0. Thus, the secular equation $\det(H_0 - E_0) = 0$ determines the value of E_0 as a pole of f_0 [8],

$$a_0 - E_0 - b_0^2 f_1(E_0) = 0 . \tag{17}$$

This may be interpreted either (simply) as a definition of the parameter a_0 or (numerically) as a transcendental equation for E_0. The latter case represents merely a continued-fractional reformulation of the Lanczos standard numerical eigenvalue method [9]. In both cases, the evaluation of a_0 or E_0 generates also the auxiliary sequence $f_1(E_0), f_2(E_0), \ldots$ as a byproduct.

(4) *An example of analysis of the continued-fractional convergence.* We have the choice of H_0 under our control. Hence, a convergence of the functions defined by (12) in the limit $N\to\infty$ may be easily guaranteed by the simple criterion [6]

$$a_n^2 > 4b_n^2 , \quad n > N_0 . \tag{18}$$

This may be best illustrated on our example (4) with a scaling freedom $r^2 \to r^2/\rho$ of the "spring constant" in the harmonic oscillator basis $|n\rangle$ [5]. The requirement $\langle n|H_1|n\rangle = 0$ fixes then the matrix elements,

$$a_n = \langle n|H|n\rangle = (\rho + \mu/\rho)\gamma_n + \lambda(\beta_{n-1}^2 + \gamma_n^2 + \beta_n^2) ,$$

$$\gamma_n = 2n + l + \tfrac{3}{2} , \quad \beta_n = (n+1)^{1/2}(n+l+\tfrac{3}{2})^{1/2} , \quad \lambda = \nu/\rho^2 , \quad n, l = 0,1,\dots , \tag{19}$$

and we may choose also

$$\langle n|H_1|n+2\rangle = \langle n+2|H_1|n\rangle = \beta_n\beta_{n+1} , \quad \langle n|H_1|n+1\rangle = \langle n+1|H_1|n\rangle = \beta_n\gamma_{n+1} ,$$

$$b_n = (-\rho + \mu/\rho + \lambda\gamma_n)\beta_n , \quad m > N_0 . \tag{20}$$

Such a choice of H_0 (5) will imply that [10]

$$b_n f_{n+1} \doteq \tfrac{1}{2}(3 - \sqrt{5}) \doteq 0.382 , \quad \nu n \gg 1 , \tag{21}$$

i.e., the continued fractions converge and decrease quickly enough to suppress also the asymptotics of the auxiliary factors d_n and g_n for $\nu n \gg 1$.

In conclusion, we may notice that the present formalism may be also combined with the ordinary RS expansions. For example, we may choose the particular scaling $\rho = \rho(M)$ in (20) such that $b_M = 0$, i.e.,

$$\rho^3(M) - \mu\rho(M) - \nu\gamma_M = 0 . \tag{22}$$

Then, the auxiliary continued fractions will terminate, and the low-order formulae acquire a finite-dimensional character resembling the RS formalism with linear combinations of the oscillator states $|n\rangle$. In the simplest cases, we even get the compact prescriptions

$$E = a_0 + \mathrm{O}(\nu^2) , \quad M = 0 ,$$

$$E = \tfrac{1}{2}\{a_0 + a_1 - [(a_1 - a_0)^2 + 4b_0^2]^{1/2}\} + \mathrm{O}(\nu^2) , \quad M = 1 , \tag{23}$$

reproducing and improving the first-order RS ground state energies, respectively. In the higher orders, a more detailed account of applications of the present formalism will be published in the near future [11].

Note added

I am obliged to the referee for noticing that the condition $b_M = 0$ is closely related to some recent analyses of optimality of the diagonal RS operators H_0. Indeed, if we treat the scaling ρ as a free parameter and try to improve the RS convergence properties by its appropriate choice ($\rho \sim \rho_{\text{optimal}}$), then the value of ρ_{optimal} is believed to be order dependent [12]. In this context, our explicit and order-dependent prescription (22) might represent a very natural tentative candidate for ρ_{optimal}.

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